

13.05 Arithmetic Series

Practice problems

Show your work and state restrictions or excluded values when needed.

Recognize	Apply these rules
Sequences And Series	Check whether the pattern is additive or multiplicative before choosing a formula.
The expression contains the chapter's key notation or model.	Write the known quantities, choose the matching formula, calculate carefully, and check the result in context.

A. Practice: Arithmetic Series

1.

$$a_{10} = 5 + (10 - 1)3$$

2.

$$a_n = 4(2)^{n-1}$$

3.

$$S_{20} = \frac{20(4 + 61)}{2}$$

4.

$$a_{10} = 5 + (10 - 1)3$$

5.

$$a_n = 4(2)^{n-1}$$

6.

$$S_{20} = \frac{20(4 + 61)}{2}$$

7.

$$a_{10} = 5 + (10 - 1)3$$

8.

$$a_n = 4(2)^{n-1}$$

9.

$$S_{20} = \frac{20(4 + 61)}{2}$$

10.

$$a_{10} = 5 + (10 - 1)3$$

11.

$$a_n = 4(2)^{n-1}$$

12.

$$S_{20} = \frac{20(4 + 61)}{2}$$

13.

$$a_{10} = 5 + (10 - 1)3$$

14.

$$a_n = 4(2)^{n-1}$$

15.

$$S_{20} = \frac{20(4 + 61)}{2}$$

16.

$$a_{10} = 5 + (10 - 1)3$$

17.

$$a_n = 4(2)^{n-1}$$

18.

$$S_{20} = \frac{20(4 + 61)}{2}$$

19.

$$a_{10} = 5 + (10 - 1)3$$

20.

$$a_n = 4(2)^{n-1}$$

B. Apply the key formulas

1.

$$a_n = a_1 + (n - 1)d$$

2.

$$a_n = a_1 r^{n-1}$$

3.

$$S_n = \frac{n(a_1 + a_n)}{2}$$

4.

$$a_n = a_1 + (n-1)d$$

5.

$$a_n = a_1 r^{n-1}$$

6.

$$S_n = \frac{n(a_1 + a_n)}{2}$$

7.

$$a_n = a_1 + (n-1)d$$

8.

$$a_n = a_1 r^{n-1}$$

9.

$$S_n = \frac{n(a_1 + a_n)}{2}$$

10.

$$a_n = a_1 + (n-1)d$$

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$$S_n = \frac{n(a_1 + a_n)}{2}$$

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$$a_n = a_1 + (n-1)d$$

20.

$$a_n = a_1 r^{n-1}$$